

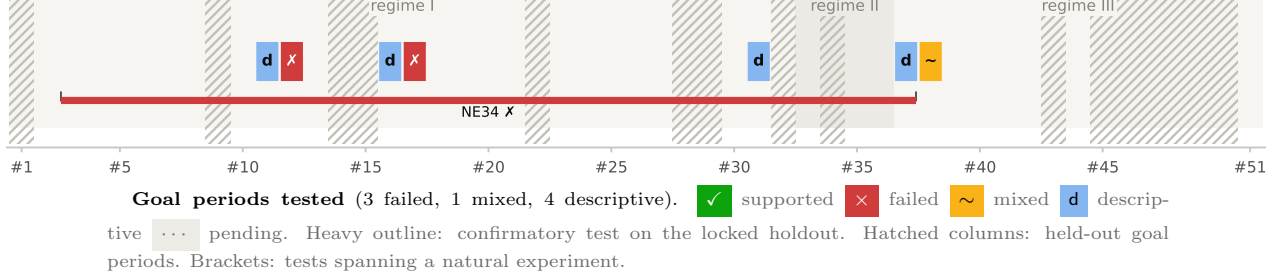
H10: Goals are Legendre pushes: a swarm's response to an assigned goal is predictable from its unforced fluctuations

Vivian Rogers, with Claude Opus 5.5 · agent-swarm physics · Updated 2026-10-03

Complete 40% Faithfulness 1/5 Usefulness 1.5/5

Bottom line. In the AI Village, goals are quenches toward a common target, not Legendre pushes: free-week fluctuations do not predict who follows a goal, and the variance moves the wrong way.

level: hypothesis · pipeline checks 86% · updated 2026-10-03



WHAT HAPPENED

Can a swarm's unforced fluctuations predict how it lines up behind an assigned goal? Each agent's 30-min statement embeddings were projected on the goal direction $\hat{g}$ (goal plus kickoff text). In three free $\rightarrow$ assigned pairs (#11 $\rightarrow$ #12, #16 $\rightarrow$ #17, #37 $\rightarrow$ #38, non-holdout), one field λ was fitted to the mean shift and everything else predicted. Alignment rose in every pair, by 1.4–3.1 free-week SDs, outside linear response, so the variance test was untestable. **The core test failed:** free-week fluctuations did not predict who moved ($r = -0.37, -0.69, +0.20$; Stouffer $p = 0.89$; synthetic power 95%). Variance along $\hat{g}$ instead grew $\times 1.9$ – 7.7 while transverse variance held, the opposite of a saturating tilt. Across 18 regime-I goal changes the jump fell with prior fluctuation (Spearman -0.56 , $p = 0.015$). Agents converge on a common alignment and switch on- and off-goal together. Holdout (#22 $\rightarrow$ #23, #31 $\rightarrow$ #32) frozen, not run.

PHYSICAL MODEL

Model 11, mean-field $O(n)$ in $n = 32$ whitened embedding dimensions. $\mathbf{s}_i$: agent i 's unit statement vectors; $y = \hat{\mathbf{z}} \cdot \hat{g}$, alignment with the goal field direction; x_i , its 30-min window mean, with free-week mean μ_i , noise-corrected variance $\kappa_{2,i}$, third cumulant $\kappa_{3,i}$. The goal is a linear field $\lambda = \beta h$ along $\hat{g}$; K is the cumulant generating function; J_0 the mean-field coupling; g the loop gain; C_F the free-week window covariance.

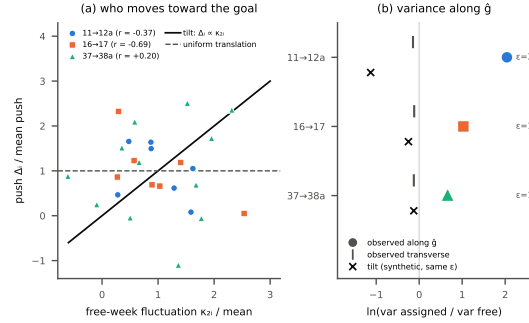
$$P_A(\mathbf{s}) \propto P_F(\mathbf{s}) e^{\lambda \sum_i \hat{g} \cdot \mathbf{s}_i}, \quad G = F - \lambda m$$

$$K_A(t) = K_F(t + \lambda) - K_F(\lambda) \Rightarrow \kappa_n^A = K_F^{(n)}(\lambda)$$

$$\Delta \mu_i = \lambda \kappa_{2,i} + \frac{\lambda^2}{2} \kappa_{3,i}, \quad \kappa_{2,i}^A = \kappa_{2,i} + \lambda \kappa_{3,i}$$

$$\chi_{\text{swarm}} = \chi_0 / (1 - \beta J_0 \chi_0), \quad g = \beta J_0 \chi_0 = 1 - V_{\text{indep}} / V$$

$$\Delta \mu = \lambda C_F \hat{g}$$



Observables. (a) Each agent's push toward the assigned goal vs. its free-week fluctuation along $\hat{g}$, both divided by the pair mean. The tilt predicts the solid line, uniform translation the dashed one; the regime-I pairs slope the wrong way. (b) Change in variance along $\hat{g}$ (colored), along transverse directions (bars), and for an exact tilt of a bounded mean-field swarm at the same push ($\times$, synthetic). The goal disperses the swarm along $\hat{g}$ instead of narrowing it.

SO WHAT? PRACTICAL VALUE FOR UNDERSTANDING AGENT SWARMS

Mostly a negative result, useful only in a narrow sense. The appealing promise, reading how steerable a swarm is from how it fluctuates when nobody steers it, does not hold here. Goal changes are huge kicks (median about 2.5 SDs), not small fields, and the agents that wander most are not the ones that follow most. What an operator learns: an assigned goal acts like a target everyone converges on and a new mode the swarm toggles collectively, not a gentle tilt of existing preferences. So pre-goal behavior is a poor forecast of compliance. The instrument (goal-direction projection, noise-corrected variance) is cheap on any agent log, but the fluctuation-response physics it was built to test is, in this village, a curiosity.

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RESULTS VS PREDICTIONS

Pending.

FAITHFULNESS SCORECARD

A mapping	1	F identifiability	1
B assumptions	1	G ground truth	1
C adequacy	0	H comparative	0
D unfitted predictions	0	I transfer	0
E interventional	0		
Level: hypothesis . Scale 0 not done or failed, 1 partial, 2 passed (<code>writeup/paper.tex</code>).			

CAVEATS

Pending.

ROUND 2

GOAL PERIODS IN DETAIL

- G11 d descriptive (F1 $\times$ by BIC, F2 $\times$ marginal, F3 $\times$ drift)
- G12 $\times$ failed (pair test #11 $\rightarrow$ #12a; NE34)
- G16 d descriptive (F1 $\times$ by BIC, F2 $\approx$ 0.5, F3 $\checkmark$)
- G17 $\times$ failed (pair test #16 $\rightarrow$ #17; NE34)
- G31 d descriptive (F1 mixed, F2 $\times$, F3 $\checkmark$)
- G37 d descriptive (F1 $\checkmark$, F2 $\times$, F3 $\checkmark$)
- G38 $\sim$ mixed (pair test #37 $\rightarrow$ #38a; NE34)
- NE34 $\times$ failed (P1 fails in combination: r $\not\leq$ 0 in 1/3 pairs, Stouffer p = 0.89; P2 untestable, every...

- **Essence.** How does a new goal propagate: as a field acting on everyone at once, or as contagion from early adopters?
- **H10-R1** Adoption onsets after a kickoff: a field predicts simultaneous onsets; contagion predicts onsets ordered by exposure to early adopters.
- **H10-R2** Goals act on a two-state on-goal/off-goal occupancy (HH130).
- **H10-R3** During the week, goal-specific content relaxes back toward family-prior directions (E4).

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